

```
In [5]: from sentence_transformers import SentenceTransformer, util
        from sentence_transformers.util import paraphrase_mining
        model = SentenceTransformer('all-MiniLM-L6-v2')
```

Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
 WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.

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```
In [6]: s1="The is one the mat"
        s2="A cat sitting on mat"
        emb1=model.encode(s1)
        emb2=model.encode(s2)
        cos_sim=util.cos_sim(emb1,emb2)
        print(cos_sim)
```

```
tensor([[0.4667]])
```

```
In [8]: sentences = [  
    "The cat sits outside",  
    "A man is playing guitar",  
    "I love pasta",  
    "The new movie is awesome",  
    "The cat plays in the garden",  
    "A woman watches TV",  
    "The new movie is so great",  
    "Do you like pizza?",  
    "She enjoys reading books",  
    "He is walking his dog",  
    "The weather is nice today",  
    "Let's go to the beach",  
    "I need a cup of coffee",  
    "They are planning a trip",  
    "She cooks delicious meals",  
    "I am learning to play the piano",  
    "He works at a tech company",  
    "This book is very interesting",  
    "The sun is shining brightly",  
    "It's raining outside",  
    "The children are playing soccer",  
    "He is fixing the car",  
    "She writes beautiful poetry",  
    "The flowers are blooming",  
    "I enjoy hiking in the mountains",  
    "He is swimming in the pool",  
    "We are watching a comedy show",  
    "The dog is barking loudly",  
    "She is painting a landscape",  
    "He likes to play video games",  
    "I am studying for my exams",  
    "They are building a new house",  
    "The coffee shop is busy",  
    "He listens to classical music",  
    "She is designing a website",  
    "I bought a new laptop",  
    "They are baking cookies",  
    "The bird sings in the morning",  
    "He loves going to concerts",  
    "She practices yoga every day",  
    "The phone is ringing",  
    "I need to finish this project",  
    "The restaurant has great food",  
    "He collects rare coins",  
    "She enjoys watching documentaries",  
    "They are exploring the city",  
    "The laptop is charging",  
    "He is training for a marathon",  
    "I meditate every morning",  
    "The bakery sells fresh bread",  
    "She is learning French"  
]
```

```
In [7]: paraphasers=paraphrase_mining(model,sentences)
paraphasers
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```
In [12]: for i in paraphasers [0:20]:
        score,i,j = i
        print(f"{score:.2f}\t\t{sentesences[i]}\t\t\t{sentesences[j]}")
```

0.89	The new movie is awesome	
	The new movie is so great	
0.68	The cat sits outside	
	The cat plays in the garden	
0.63	She enjoys reading books	
	She enjoys watching documentaries	
0.58	They are planning a trip	
	They are exploring the city	
0.58	He listens to classical music	
	He loves going to concerts	
0.53	I need a cup of coffee	
	The coffee shop is busy	
0.52	He likes to play video games	
	He loves going to concerts	
0.52	I bought a new laptop	
	The laptop is charging	
0.51	I love pasta	Do you
	like pizza?	
0.50	She enjoys reading books	
	She is learning French	
0.50	She cooks delicious meals	
	The restaurant has great food	
0.48	He likes to play video games	
	He listens to classical music	
0.48	I am learning to play the piano	
	I am studying for my exams	
0.48	She writes beautiful poetry	
	She is painting a landscape	
0.48	The weather is nice today	
	The sun is shining brightly	
0.47	A woman watches TV	
	She enjoys watching documentaries	
0.47	She cooks delicious meals	
	She writes beautiful poetry	
0.47	She is painting a landscape	
	She is designing a website	
0.47	They are baking cookies	
	The bakery sells fresh bread	
0.45	She enjoys reading books	
	She writes beautiful poetry	

In [13]: `model = SentenceTransformer("distiluse-base-multilingual-cased-v1")`

```
modules.json:  0%|          | 0.00/341 [00:00<?, ?B/s]

config_sentence_transformers.json:  0%|          | 0.00/122
[00:00<?, ?B/s]

README.md:  0%|          | 0.00/2.24k [00:00<?, ?B/s]

sentence_bert_config.json:  0%|          | 0.00/53.0 [00:00<?, ?
B/s]

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model.safetensors:  0%|          | 0.00/539M [00:00<?, ?B/s]

Loading weights:  0%|          | 0/100 [00:00<?, ?it/s]

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config.json:  0%|          | 0.00/190 [00:00<?, ?B/s]

config.json:  0%|          | 0.00/114 [00:00<?, ?B/s]

2_Dense/model.safetensors:  0%|          | 0.00/1.58M [00:00<?, ?
B/s]
```

In [28]: `corpus1 = ["The future is mobile.",
"The future is digital."
"The future is electric."
"ipem college lies in uttar pradesh"]
corpus_emb = model.encode(corpus1)`

```
In [29]: # Computer pairwise cosine similarity
cos_similarity = util.pytorch_cos_sim(corpus_emb,corpus_emb)
avg_sim = cos_similarity.mean(dim=1)
outlier_index = avg_sim.argmin().item()
outlier_index
print(f"Outlier sentence : {corpus1[outlier_index]} (AVG Similarity :
{avg_sim[outlier_index]}")
```

Outlier sentence : The future is mobile. (AVG Similarity : 0.749106764793396)

```
In [34]: corpus1 = ["I have a nice dog.",
                    "my dog is a well trained pet.",
                    "Instagram is a product of META."]
corpus_emb = model.encode(corpus1)
```

```
In [35]: # Computer pairwise cosine similarity
cos_similarity = util.pytorch_cos_sim(corpus_emb,corpus_emb)
avg_sim = cos_similarity.mean(dim=1)
outlier_index = avg_sim.argmin().item()
outlier_index
print(f"Outlier sentence : {corpus1[outlier_index]} (AVG Similarity :
{avg_sim[outlier_index]}")
```

Outlier sentence : Instagram is a product of META. (AVG Similarity : 0.41154971718788147)

```
In [36]: corpus1 = ["Cats are very playful animals.",
                    "Dogs are loyal pets.",
                    "Rabbits love eating carrots.",
                    "Artificial Intelligence is transforming industries."]
corpus_emb = model.encode(corpus1)
```

```
In [37]: # Computer pairwise cosine similarity
cos_similarity = util.pytorch_cos_sim(corpus_emb,corpus_emb)
avg_sim = cos_similarity.mean(dim=1)
outlier_index = avg_sim.argmin().item()
outlier_index
print(f"Outlier sentence : {corpus1[outlier_index]} (AVG Similarity :
{avg_sim[outlier_index]}")
```

Outlier sentence : Artificial Intelligence is transforming industries. (AVG Similarity : 0.329438716173172)

In []: